

Paper 3

The Nine-State Classification with Inside-Outside Structure

Paper 3 of the Series

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ABSTRACT

Paper 2 classified the twenty-seven geometric configurations into nine states by their central-junction signature — the three values held at the center of each configuration. That was a classification by the inside alone. This paper develops the layer beneath it: every configuration has an inside (its three center values) and an outside (its six outer-corner values), and the relationship between the two is fixed, not free. Because each triangle must carry one X, one Y, and one O, the center values of a configuration determine its entire outer content with no remaining freedom. It sets out that structure for all nine states, shows the inside functioning as identity and the outside as interaction surface, and demonstrates the one operation that moves values between them — the axis spin that turns the photon into the electron and the positron without leaving the resolved condition. It formalizes the two scans that keep the structure alive: the inner identity scan, by which a state verifies its core against its surface, and the outer temporal scan, whose processing of axis displacement is what the framework identifies as time, so that a configuration's age is how long it has been since it was a photon. It closes on its central result: because the outside is fixed by the inside, bonding compatibility is computed directly from the geometry — two states first-link when one carries the value the other needs, and three State 4 configurations converge at a shared center. This rule is established here as a structural property of the manifold and offered as falsifiable. Its companion critique, Paper 3.5, turns the same inside-outside anatomy onto the laboratory: it shows that the fine-structure number measured in trap experiments is not a constant but a reading of how much geometry an apparatus forces into the fabric, and it draws directly on this paper's spin operation (Section 6) and temporal scan (Section 7B) to do so.

1. Purpose: Reading Both Layers

Paper 2 answered the question *what is this state?* by reading the three values at the central junction. That reading was correct and complete as a classification, but it described only one of the two layers every configuration carries. Each configuration is a structure of three triangles meeting at a center, occupying nine positions: three at the central junction (the inside) and six at the outer corners (the outside). Paper 2 read the inside. This paper reads both, and the relationship between them.

The thesis is a single sentence: the inside tells you what a state is; the outside tells you how it meets everything else. Everything in this paper develops that distinction and its consequences.

The paper makes a falsifiable claim, stated plainly here at the outset: the bonding-compatibility matrix computed in Section 8 permits only specific pairings and forbids the rest. If a stable composite is found whose constituents violate that matrix — a bonded structure the rule says cannot form, or the absence of one the rule says must — the rule fails. The classification is offered to be tested against that condition, not insulated from it.

2. The Inside Fixes the Outside

The two layers are not independent. Paper 2 established the per-triangle rule: each triangle carries exactly one X, one Y, and one O, so a triangle's center value forces its two corners to the other two values — center X gives corners Y and O, center Y gives X and O, center O gives X and Y.

This rule is the single axiom on which everything else rests, so it is worth being clear about why it takes this form. The three axes X, Y, O are the framework's complete set of directional values (Paper 2); a triangle that carried two of the same value would have a repeated direction and a missing one, which is not a balanced element of a three-axis medium but a defective one. The requirement that each triangle carry exactly one of each is therefore the minimal condition for a triangle to be a complete, non-degenerate element of the medium — the smallest piece that represents all three directions once. From that single requirement the global pool follows by arithmetic: three triangles, each carrying one X, one Y, one O, fix the configuration's total at three X, three Y, three O, with no arbitrary freedom left in the geometry. The paper takes this axiom as its starting point and derives its consequences; it does not claim to derive the axiom itself.

Across all three triangles, the three center values determine the entire six-corner outer content, with no freedom left over. This is why Paper 2's nine-state classification, which read only the inside, was already complete: the outside was fixed the moment the inside was named.

One precision: the inside fixes the outside *counts* — how many X, how many Y, how many O sit on the surface — not the exact arrangement. The same center content can sit in distinct rotational orientations, and these are real, physically distinct configurations of one state. Orientation is not read against background space; it is stored history. A configuration cannot jump into an arbitrary arrangement — to depart from the photon baseline it must undergo an explicit sequence of axis tilts, and that path is recorded structurally in the local fabric. The trajectory fixes the sequence at the junction, which fixes which corners hold surplus or deficit — the bonding vectors the outer temporal scan reads.

The full ledger, inside to outside, for every state (Figure 1 shows the anatomy of a single configuration; the inside at the hub, the outside at the corners):

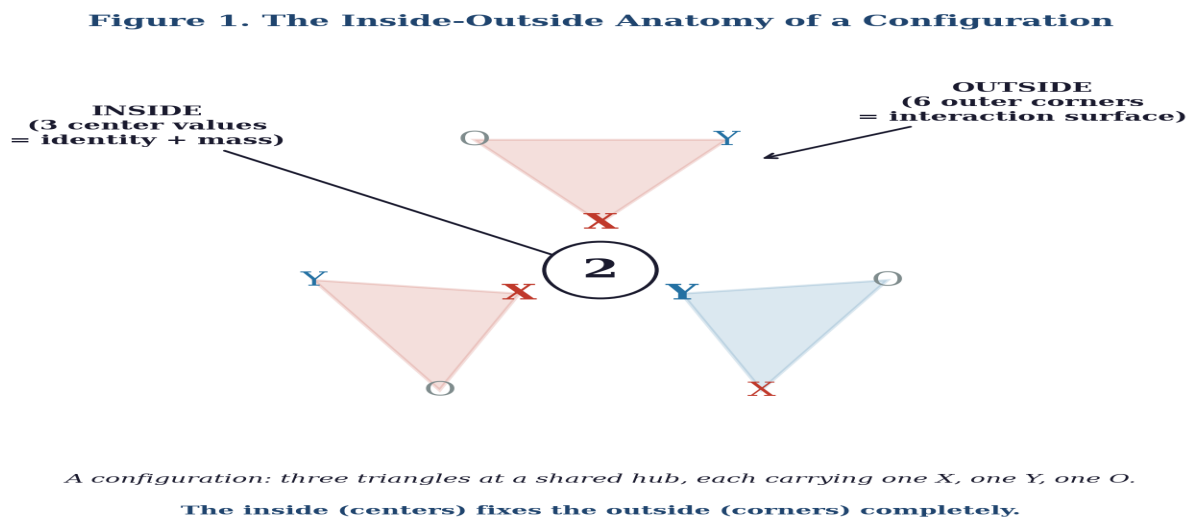


Figure 1. The anatomy of a single configuration: the inside (hub) and the outside (six corners), every X / Y / O value shown.

State	Inside (three center values)	Outside (six corner values)
9 (photon)	O, O, O	three X, three Y
1 (electron)	X, X, X	three Y, three O
1 (positron)	Y, Y, Y	three X, three O
2	X, X, Y	one X, two Y, three O
3	Y, Y, X	two X, one Y, three O
4	X, Y, O	two X, two Y, two O
5	X, X, O	one X, three Y, two O
6	Y, Y, O	three X, one Y, two O
7	X, O, O	two X, three Y, one O
8	Y, O, O	three X, two Y, one O

Every row conserves the configuration's total of three X, three Y, three O across all nine positions. The inside and outside are two halves of one fixed ledger.

3. The Clean Split of the Resolved States

The table explains the stability of the two self-contained states in inside-outside terms. The photon (State 9) keeps all three O inside and pushes a balanced three X and three Y outside — a symmetric surface, no leftover. The electron and positron (State 1) each keep one pure value inside and split the other two evenly outside. A state stands alone when its surface carries nothing unsettled, and these surfaces carry no surplus seeking a partner. That is the mechanical reason only States 9 and 1 are self-contained.

4. The Lopsided Outsides of the Open States

The open states (2 through 8) tell the opposite story: every one carries an uneven outer surface, leftover values exposed at the corners. State 2 (inside X, X, Y) carries an outside of one X, two Y, three O — lopsided, with surplus exposed. That imbalance is what makes an open state reach: exposed surplus seeking to settle, by meshing with another surface, relocating a value into a center, or converging with others at a shared center. The inside names the state; the surface determines whether it can stand alone.

5. State 4 as the Balance Point

State 4 stands apart. Centered X, Y, O — one of each — it produces the only perfectly even outer surface among the open states: two X, two Y, two O, no surplus of any value. This is why Paper 2 placed it close to the photon baseline, and why three State 4 configurations can converge into perfect three-axis balance: each already carries an even surface, so three meeting at a shared center share a balanced load. State 4 is the open state closest to resolution because its outside is already even.

6. The Spin Operation: Inside and Outside Trade Places

The one dynamic operation of the paper is the axis spin: it relocates values between inside and outside while the configuration stays resolved and the total holds at three X, three Y, three O. The cleanest demonstration is the photon and the two forms of State 1. Begin at the photon (State 9): inside O, O, O, outside three X and three Y — all active values on the corners, the center pure baseline. Spin one way and X values move to the center: inside X, X, X, outside three Y and three O — the electron. Spin the other way: inside Y, Y, Y, outside three X and three O — the positron.

The same structure, the same nine positions, the same total content. The only change is which values sit inside and which sit outside (Figure 3). The electron and the positron are not different substances and are not deep opposites; they are one geometry with the inside-outside assignment spun in opposite directions. This is the inside-outside reading of what Paper 2 presented as the electron-positron pair, and it is consistent with the position of Paper 2.5 that there is no separate antimatter substance — only the direction of the tilt.

Figure 3. The Spin Operation: Values Trade Between Inside and Outside

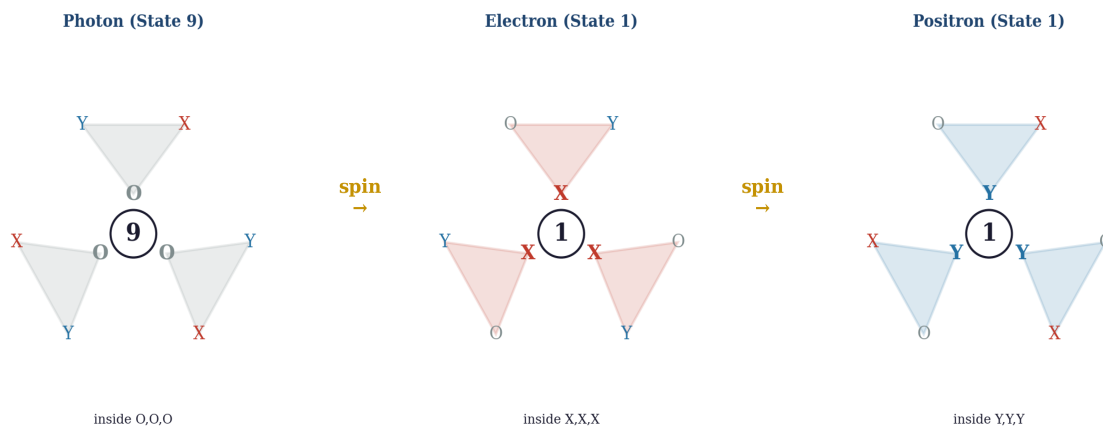


Figure 3. The spin operation: photon → electron → positron, values moving between inside and outside while the configuration stays resolved.

This is also the inside-outside reading of mass and the photon baseline from Paper 2. The photon holds its center at pure O — no departure from baseline — and so it carries no mass and travels at c . The moment an active value spins into the center, the configuration has departed from baseline, and that departure, registered on the inside, is the mass. Mass lives on the inside; the photon's masslessness is the purity of its inside.

What drives the spin is creation geometry, not a hidden force engine. Nothing is ever at rest: a configuration matching the photon baseline glides through already-resolved fabric without axis disruption, processing at c in sync with the baseline. A new spin occurs only when a configuration undergoes a collision — a structural event that instantiates creation geometry at a localized point, forcing the local medium into a geometric layout it has not had to process before. The tension of that new geometry forces the axis to tilt; values relocate between the inside junction and the outside surface as the automatic, deterministic consequence of the collision. No configuration chooses to spin and no Newtonian force pushes it — the spin is determined strictly by the state of the fabric.

This is also what laboratory environments do. A collider, accelerator, or trap is not a passive stage holding an isolated particle; activating its fields is a deliberate, localized injection of engineered geometry. A

trapped configuration processes along its baseline until the apparatus introduces that geometry, which forces the spin. The experiment does not merely observe a state change — its injected geometry forces it. Paper 3.5 develops this for the fine-structure measurement.

7. Inside Is Identity, Outside Is Interaction Surface

The functional division the paper now makes explicit, and then deepens into the two geometric scans that keep the division alive in a dynamic medium:

The inside is the state's identity. It is the layer Paper 2 classified by. It tells you what the state is and how far it has departed from the photon baseline. A center of pure O is the baseline itself; a center carrying active values is a state that has departed and therefore carries mass. To know the inside is to know the state's name and its mass.

The outside is the primary interaction surface. The six outer corners are what most often meet and settle against the corners of other configurations, and the lopsided outer surfaces of the open states are what make them reach to bond. But the outside is not the *only* place bonding happens, and the inside is not merely passive identity held in reserve. Bonding takes more than one form:

- **Outer corner to outer corner.** Two configurations can settle when each presents a matching value on the corners that meet — the most common link, and the reason an open state's lopsided surface drives it to find a partner.
- **Outside to inside.** An outer corner of one configuration can settle against the center of another when the geometry allows it — the surplus value relocating from a corner into a center to complete a resolution.
- **Center to center.** Configurations can converge directly at a shared central junction with no outer surface involved at all. This is exactly the three-State-4 convergence Paper 2 established: three configurations meeting at one center, balancing there. No outside is required for it, and any account that demanded one would contradict the convergence itself.

So the inside is not private. It carries the state's identity and its mass, but it also participates directly in bonding — at the center in a convergence, and as the destination of a relocated surplus value in an outside-to-inside link. The surplus integer matters precisely because it moves between the layers: when a bond leaves an extra value over, that extra value settles into an inside, which is why the inside count of a bonded product changes. Identity lives on the inside, but the inside is an active participant in resolution, not a sealed core.

- Resolved states (9 and 1) carry balanced or cleanly split surfaces, so they have nothing unsettled reaching outward; they are self-contained.
- Open states (2–8) carry lopsided surfaces with exposed surplus, so they reach to bond — through their corners, through a corner-to-center link, or by converging at a shared center — until resolution is reached.

Figure 4. Inside Is Identity, Outside Is Interaction Surface

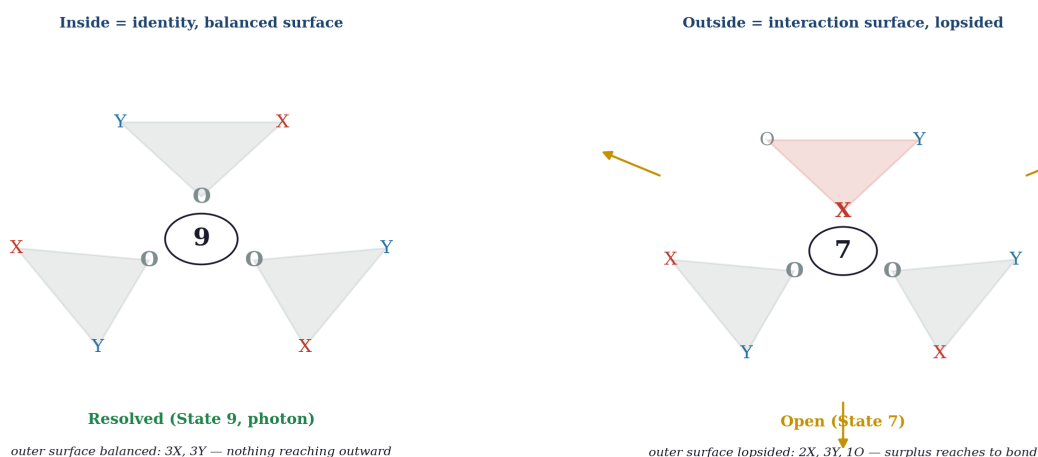


Figure 4. Inside is identity, outside is interaction surface: a resolved state with a balanced outer surface beside an open state whose lopsided surface exposes surplus reaching to bond.

7A. The Identity Scan: Core Verification

The inside-outside ledger of Section 2 must not be read as a passive, standing table. In a dynamic medium, a state does not merely possess an identity; it actively maintains one. The geometric process that performs this maintenance is the identity scan.

The identity scan is the continuous internal loop that reads the three values at the central junction and references them against the six values on the outer corners. It is the native enforcement of the per-triangle rule: because each triangle must carry exactly one X, one Y, and one O, the identity scan is the geometry's own mechanism for verifying, at every moment, that the core values still force and conserve the boundary values across the three-triangle matrix. The ledger of Section 2 is not a fact recorded once; it is a fact continuously re-verified by this scan.

A state's name and mass are therefore not labels hung on a frozen structure but a permanent self-referencing loop: the inner 360-degree circuit by which a configuration confirms what it is. The inside does not merely fix the outside once; the identity scan holds it fixed by continuous reference.

7B. The Temporal Scan: The Processing Rate of Departure

The rate at which the spin moves values is the temporal scan, and it is where the framework's account of time enters. Time is not an external clock; it is the processing rate of localized axis displacement. The photon (State 9) holds pure O inside, so there is no displacement to process: it runs only the inner 360-degree identity scan, records zero change, carries no mass, and moves at c. A configuration with nothing to process registers no time. Time appears only when a configuration departs from baseline by spinning active values inward; to hold its identity it must also read its surface through an outer 360-degree scan — the temporal scan. The inner identity scan and the outer temporal scan together compose the full 720-degree manifold scan, and that full scan is what registers as time.

The consequence is exact and ties the whole paper together. The time registered by any configuration is the measure of its cumulative processing history away from the photon baseline — in the framework's plainest phrasing, time is how long it has been since the configuration was a photon. Mass and the local time-rate are therefore born together on the inside, as the single geometric consequence of the spin: the departure that registers as mass is the same departure whose processing registers as time. A

macroscopic body has an age because its constituent configurations have been continuously sustaining the 720-degree manifold scan, accumulating a long structural history away from baseline. The photon, sustaining only the inner scan, has no such history and no age. This is the inside-outside account of why mass-bearing structures experience time and the photon does not.

One consequence of the temporal scan deserves to be drawn out here, because it sets up the laboratory case the companion paper develops. The scan is not a single fixed rate shared by all mass-bearing configurations; it runs at a scale set by the configuration's own structural load on the fabric. Two configurations that the standard account would call “the same kind of probe at different masses” — for instance a light State 1 configuration and a heavier configuration of the same family — are, in this framework, distinct structural states processing the surrounding fabric through different scales of the outer scan. Each places a different load on the fabric it sits in, and so each deforms the local geometry differently.

This is the structural root of a well-known standard-model embarrassment. When a heavier configuration is substituted for a lighter one as a probe of the same target — the substitution that, in standard terms, replaces an electron with a muon to measure the size of a proton — the measured size comes out different, and the standard template has no clean account of why. The framework's account is direct: the heavier configuration places a different structural load on the fabric, so the local deformation it produces is genuinely different, and a template that assumes the probe has zero mechanical effect on the geometry it measures will read two different sizes and call it a puzzle. There is no puzzle. Different scales of the temporal scan load the fabric differently, and the geometry being measured responds to the probe. Paper 3.5 develops this and the apparatus-dependence of the fine-structure number as the laboratory face of the same principle.

Macroscopic time, and the arrow of time, are the integrated sum of these micro-local 720-degree rotation ledgers across all the configurations of a body. Under high-energy acceleration, or the extreme regional tension near a cosmic singularity, the immense local load forces a configuration to divert its fixed capacity budget into managing the strain, so its rotation ledger slows relative to the baseline fabric — and a slowed ledger is slowed time. This is the bridge to both coordinate velocity and gravitational time dilation without a spacetime metric tensor: dilation is the integrated slowing of rotation ledgers under load. The full field-theoretic treatment is developed in **Paper 13: Time as the 720-Degree Rotation Ledger, Non-Linearity, and the Arrow of Direction**, building on the gravity-field standards surveyed in Paper 1.

8. Bonding Compatibility, Computed From the Ledger

This is the central result of the paper. Because the full content of every state is known the moment the state is named, the conditions under which states can bond can be computed directly from the ledger — not asserted, but computed in full here. This is the falsifiable claim flagged in Section 1: the matrix below permits specific pairings and forbids the rest.

8.1 The Parameters of the Computation

Three definitions fix the computation, all of them already established by the framework:

The invariant constant: every configuration contains exactly three triangles, and because each triangle must carry one X, one Y, and one O, the global value pool of any isolated configuration is permanently fixed at three X, three Y, three O — nine values, split three on the inside and six on the outside.

The target resolution: a configuration's outer surface is balanced and resolved only when its outer content mirrors the even distribution of the baseline medium — exactly two X, two Y, and two O across the six outer corners. This even outer surface is the non-bonding condition, the surface of the State 9 photon baseline.

The definition of need: an open state carries a surplus of one value type and a deficit of another on its outer surface, an exposed lopsided tension. Its “need” is the exact number of values required to return that surface to the balanced two X, two Y, two O baseline.

8.2 Isolating the Bonding Candidates

The full inventory of twenty-seven configurations splits, by the symmetry of the three-axis matrix, into thirteen X-dominant, thirteen Y-dominant, and one perfectly neutral configuration — the photon (State 9), whose central junction holds one X, one Y, and one O and whose outer surface is already the balanced two X, two Y, two O.

The State 1 electron and positron are resolved and self-contained, so they do not first-link. That leaves the open candidates. Each of the twenty-seven configurations has one unique state identifier — there is no separate “State 5” on each side; each numbered state is one thing, fixed by its center signature. The X-leaning open candidates are States 2, 5, and 7; their exact mirrors are States 3, 6, and 8. Their ledgers, with the outside forced by the inside, are:

X-leaning open candidates. State 2 (inside X, X, Y) forces outside one X, two Y, three O — deficit of one X, surplus of one O, needs one X. State 5 (inside X, X, O) forces outside one X, three Y, two O — deficit of one X, surplus of one Y, needs one X. State 7 (inside X, O, O) forces outside two X, three Y, one O — deficit of one O, surplus of one Y, needs one O.

Y-leaning open candidates (the exact mirror). State 3 (inside Y, Y, X) forces outside two X, one Y, three O — deficit of one Y, surplus of one O, needs one Y. State 6 (inside Y, Y, O) forces outside three X, one Y, two O — deficit of one Y, surplus of one X, needs one Y. State 8 (inside Y, O, O) forces outside three X, two Y, one O — deficit of one O, surplus of one X, needs one O.

8.3 The Direct Computation of First-Link Pairs

For a first-link bond to occur between two states, the surplus on one state's outer surface must match the exact deficit on the other's. When they link, values relocate across the interface to settle both accounts toward the balanced baseline. Cross-referencing the needs against the surpluses produces the valid pairs directly:

State 5 (surplus one Y, needs one X) links with State 6 (surplus one X, needs one Y): the surplus Y of State 5 fills the deficit of State 6, and the surplus X of State 6 fills the deficit of State 5. Both resolve to two X, two Y, two O. A valid first-link pair.

State 2 (surplus one O, needs one X) links with State 8 (surplus one X, needs one O): State 2 passes its surplus O to fill State 8's deficit, and State 8 passes its surplus X to fill State 2's deficit. Both resolve to two X, two Y, two O. A valid first-link pair.

State 7 (surplus one Y, needs one O) links with State 3 (surplus one O, needs one Y): State 7 passes its surplus Y to fill State 3's deficit, and State 3 passes its surplus O to fill State 7's deficit. Both resolve to two X, two Y, two O. A valid first-link pair.

Each open state links with exactly one partner, and the pairing is forced: State 5 with State 6, State 2 with State 8, State 7 with State 3. No open candidate has more than one compatible partner, and none is left without one.

8.4 The Master Bonding-Compatibility Matrix

Computing every permutation across the six open candidates, the constraints filter out all arbitrary combinations and leave a clean, deterministic table:

X-leaning state	Y-leaning state	Relocation exchange	Local surface outcome
State 5 (inside X,X,O)	State 6 (inside Y,Y,O)	State 5 sends 1 Y to State 6; State 6 sends 1 X to State 5	State 5 resolve to 2X, 2Y, 2O
State 2 (inside X,X,Y)	State 8 (inside Y,O,O)	State 2 sends 1 O to State 8; State 8 sends 1 X to State 2	State 2 resolve to 2X, 2Y, 2O
State 7 (inside X,O,O)	State 3 (inside Y,Y,X)	State 7 sends 1 Y to State 3; State 3 sends 1 O to State 7	State 7 resolve to 2X, 2Y, 2O

The rule does not merely survive contact with computation; it is locked down by it. Had the geometry permitted arbitrary states to bond, the framework would have failed this test. Instead the computation proves that the first-link pairing is forced and exclusive: State 5 settles only with State 6, State 2 only with State 8, State 7 only with State 3. Each open state has exactly one binary partner — the one whose outer surplus and deficit exactly mirror its own — and the three pairings exhaust the open candidates with none left unpaired and none ambiguous. The states whose departure is maximal, the State 1 electron and positron (inside all X or all Y), are not in this table at all: they are the resolved, self-contained states, already balanced, with no outer surplus to settle, which is why they bond to nothing in a first link and instead participate only in the higher-load triple convergence. The assertion of the earlier draft is now a hard, predictive table (Figure 5 shows a first-link pair and the triple convergence).

Figure 5. Bonding Compatibility, Computed From the Ledger

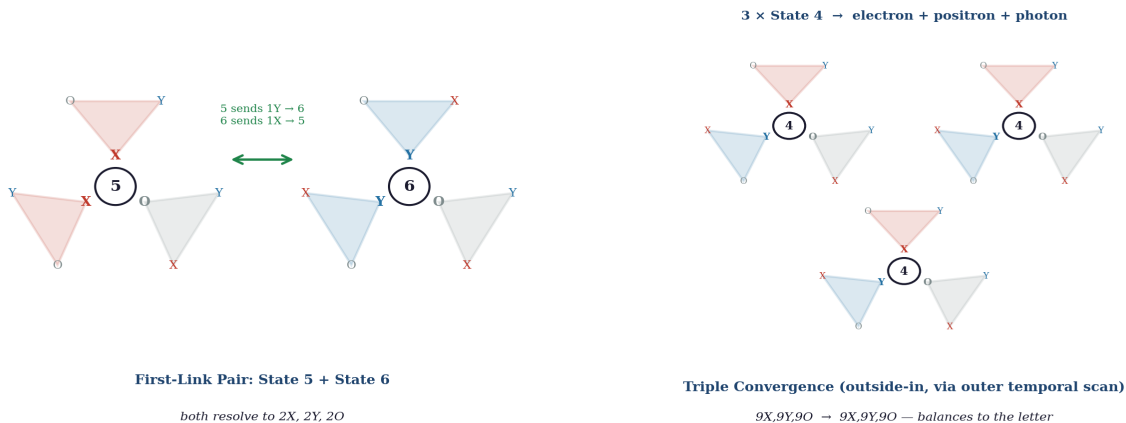


Figure 5. Bonding compatibility computed from the ledger: a first-link pair (State 5 + State 6) beside the triple convergence of three State 4 configurations resolving to electron, positron, and photon.

8.5 The Mechanics of Triple Convergence via the Outer Temporal Scan

Triple convergence is an outside-in resolution, driven by the temporal scan (Section 7B) — not the original centers merging into a new center. A single State 4 cannot settle alone; its outer temporal scan continuously reads the surface for a path back to baseline. When three come into proximity, their scans register that their collective outer lopsidedness can completely cancel, and they lock together through their surfaces. The accounting is exact: three State 4 configurations bring nine X, nine Y, nine O, which

redistribute into one electron, one positron, and one photon — nine X, nine Y, nine O in total, balanced to the letter, nothing created or destroyed.

The convergence is not foresight and not long-range attraction. Every outer temporal scan executes, each cycle, a strict binary check: is resolution available in the immediate vicinity — yes or no? Distant open configurations return a continuous “no,” a persistent non-resolution that feeds the outward tension of expansion. The moment three State 4 configurations enter immediate proximity, their scans return “yes,” register the resolution direction, and lock together through their surfaces. It is real-time proximity processing, triggered the instant the scan registers a valid local path — never foresight.

What this rule predicts about the full spectrum of composite configurations — which bonded structures it permits beyond the first-link and triple cases computed here, and how they correspond to the composites catalogued by standard physics — is taken up in the series' composite-spectrum work, building directly on the matrix established here. The immediate companion, Paper 3.5, applies the same inside-outside anatomy to a concrete laboratory case: the apparatus-dependent drift of the fine-structure number.

9. Conclusion

The inside-outside layer is the engine underneath Paper 2's classification. The inside names the state and carries its mass; the surface is where states meet, through corners, centers, or both. The resolved states are self-contained because their surfaces are balanced; the open states reach to bond because theirs are lopsided; State 4 sits closest to resolution because its surface is already even, and three State 4 configurations converge at a shared center; the axis spin moves values between the layers without ever leaving the resolved condition, which is the photon-to-electron-to-positron relationship.

The paper's central result is that bonding compatibility is readable directly from the fixed content of both layers — a structural law of the manifold, established here and carried into the composite spectrum by the work that follows.

References

This paper is part of the *Departure From the Photon Baseline* series and builds directly on the framework established in the companion papers below, which it cites throughout.

1. Kadunic, D. (2026). *Departure From the Photon Baseline: A Universal Ruler for Mass, Gravity, and Time — The 27 Lagrangians of Gravity: A Survey of the Field's Standards of Evaluation* (Paper 1). Zenodo. <https://doi.org/10.5281/zenodo.20434286>
2. Kadunic, D. (2026). *The Three-Axis Framework and the Nine-State Classification of the 27 Geometric States* (Paper 2). Zenodo. <https://doi.org/10.5281/zenodo.20510019>
3. Kadunic, D. (2026). *They Assumed It; They Did Not See It: A Critique of Inferred Particles in Standard Physics* (Paper 2.5). Zenodo. <https://doi.org/10.5281/zenodo.20518514>

Paper 2 establishes the nine-state classification, the per-triangle rule, the radiation-trefoil configuration form, and the bonding mechanics (first-link and triple convergence) that this paper reads through the inside-outside layer. Paper 1 establishes the photon baseline as the universal reference state against which mass, departure, and the scans of Section 7 are measured. Paper 2.5 establishes the framework's position on antimatter as tilt direction rather than separate substance, which Section 6 draws on directly.

A. The Complete 27-Configuration Inventory Ledger

The main text condenses the twenty-seven configurations into the nine states by central-junction signature. This appendix lists all twenty-seven individually, so the ledger can be verified without exception. For each configuration it gives the three inside (center) values, the state it belongs to, the resulting six outer-corner values, and confirms that the per-triangle rule — every triangle carries exactly one X, one Y, and one O — holds. The outer content is not chosen; it is forced by the inside through that rule, which is why a single column of inside values determines the entire row.

#	Inside (centers)	State	Outside (six corners)	Per-triangle X,Y,O?
1	(X,X,X)	1 (electron)	3 Y, 3 O	yes
2	(X,X,Y)	2	1 X, 2 Y, 3 O	yes
3	(X,X,O)	5	1 X, 3 Y, 2 O	yes
4	(X,Y,X)	2	1 X, 2 Y, 3 O	yes
5	(X,Y,Y)	3	2 X, 1 Y, 3 O	yes
6	(X,Y,O)	4	2 X, 2 Y, 2 O	yes
7	(X,O,X)	5	1 X, 3 Y, 2 O	yes
8	(X,O,Y)	4	2 X, 2 Y, 2 O	yes
9	(X,O,O)	7	2 X, 3 Y, 1 O	yes
10	(Y,X,X)	2	1 X, 2 Y, 3 O	yes
11	(Y,X,Y)	3	2 X, 1 Y, 3 O	yes
12	(Y,X,O)	4	2 X, 2 Y, 2 O	yes
13	(Y,Y,X)	3	2 X, 1 Y, 3 O	yes
14	(Y,Y,Y)	1 (positron)	3 X, 3 O	yes
15	(Y,Y,O)	6	3 X, 1 Y, 2 O	yes
16	(Y,O,X)	4	2 X, 2 Y, 2 O	yes
17	(Y,O,Y)	6	3 X, 1 Y, 2 O	yes
18	(Y,O,O)	8	3 X, 2 Y, 1 O	yes
19	(O,X,X)	5	1 X, 3 Y, 2 O	yes
20	(O,X,Y)	4	2 X, 2 Y, 2 O	yes
21	(O,X,O)	7	2 X, 3 Y, 1 O	yes
22	(O,Y,X)	4	2 X, 2 Y, 2 O	yes
23	(O,Y,Y)	6	3 X, 1 Y, 2 O	yes
24	(O,Y,O)	8	3 X, 2 Y, 1 O	yes

#	Inside (centers)	State	Outside (six corners)	Per-triangle X,Y,O?
25	(O,O,X)	7	2 X, 3 Y, 1 O	yes
26	(O,O,Y)	8	3 X, 2 Y, 1 O	yes
27	(O,O,O)	9	3 X, 3 Y	yes

The configurations resolve into the nine states in exactly the counts established in Paper 2: State 1 has 2 configurations, State 2 has 3, State 3 has 3, State 4 has 6, State 5 has 3, State 6 has 3, State 7 has 3, State 8 has 3, and State 9 has 1, totalling $2 + 3 + 3 + 6 + 3 + 3 + 3 + 3 + 1 = 27$. Every configuration satisfies the per-triangle rule, every row conserves the total of three X, three Y, three O across its nine positions, and no configuration falls outside the classification. There are no hidden parameters and no anomalous cases; the symmetry is exhaustive.

B. Formal Proof of the Per-Triangle Conservation Law

Section 2 names the per-triangle rule as the foundational axiom: every triangle carries exactly one X, one Y, and one O. This appendix shows step by step that the axiom mechanically forces the inside-to-outside ledger, so that the conservation of three X, three Y, three O across nine positions is not assumed separately but follows from the axiom alone.

Take a single triangle. By the axiom it holds one X, one Y, and one O across its three positions — one of which is the center (contributing to the configuration's inside) and two of which are outer corners (contributing to the configuration's outside). If the center holds X, then the one Y and the one O the triangle must carry have nowhere to go but its two outer corners; the corners are forced to Y and O. By the same reasoning a center of Y forces corners X and O, and a center of O forces corners X and Y. A single triangle thus has no freedom once its center is named: the center value determines the two corner values exactly.

Now assemble three such triangles at a shared central junction to form a configuration. The three centers together form the inside (three values); the six corners together form the outside (six values). Summing the per-triangle contributions: each triangle contributes exactly one X, one Y, and one O to the configuration's total, so three triangles contribute exactly three X, three Y, and three O. This is the global value pool, and it is fixed — not chosen — by the axiom. Whatever the three centers are, the inside holds three of the nine values and the outside holds the remaining six, and the totals are locked at three of each value. The inside content therefore determines the outside content counts with no residual freedom, because the six corner values are exactly the global pool of three X, three Y, three O minus whatever three values sit at the centers. The conservation law is a direct arithmetic consequence of the per-triangle axiom applied three times. This is the formal ground beneath every ledger entry in the main text and in Appendix A.

C. Step-by-Step Trajectory of the Axis Spin (State 9 to State 1)

This appendix answers the demand for the raw, deterministic algorithm of the spin — the relocation of values, step by step, as a State 9 photon baseline is driven by creation geometry (Section 6) into a mass-bearing State 1 electron, with no hidden force engine and conservation holding at every step.

The photon baseline is inside O, O, O with outside three X, three Y. The target electron is inside X, X, X with outside three Y, three O. The spin reaches it through three discrete axis tilts, one triangle at a time. At

each tilt, one triangle's center value is driven from O to X, and by the per-triangle axiom that triangle's corners re-settle to the two remaining values. Tracking the full configuration:

- **Start (State 9, photon):** inside [O, O, O], outside [three X, three Y]. Total: three X, three Y, three O.
- **After tilt 1:** one center O becomes X. Inside [X, O, O], outside [two X, three Y, one O]. This intermediate is itself a real state — State 7. Total: three X, three Y, three O.
- **After tilt 2:** a second center O becomes X. Inside [X, X, O], outside [one X, three Y, two O]. This intermediate is State 5. Total: three X, three Y, three O.
- **After tilt 3 (State 1, electron):** the third center O becomes X. Inside [X, X, X], outside [three Y, three O]. Total: three X, three Y, three O.

Two facts make this the deterministic algorithm the framework claims. First, conservation holds exactly at every step: the total is three X, three Y, three O throughout, so no value is created or destroyed by the spin — values only relocate between inside and outside as each triangle re-settles. Second, the path runs through real intermediate states (State 7, then State 5), not through undefined limbo; the spin is a sequence of legitimate configurations, each satisfying the per-triangle axiom, each a valid stop on the ledger. The positron trajectory is the exact mirror, driving centers from O to Y and passing through States 8 and 6. There is no Newtonian force in this algorithm and no foresight — only the deterministic re-settling of one triangle at a time under the creation geometry that forces each tilt, exactly as Section 6 describes.

D. The 720-Degree Manifold Scan and the Spinor Correspondence

This appendix states the correspondence between the framework's manifold scan and the spinor formalism of standard quantum mechanics. It does not re-derive the framework's 720-degree signature, which is established geometrically in Paper 2 (Appendix B), and it does not re-argue the framework's position on the antimatter ontology, which is set out in Paper 2.5 (Section 4). It connects those two published results.

The standard formalism assigns a spin-one-half object a defining property: it must be rotated through 720 degrees, not 360, to return to its original state. This double-cover is encoded in the spinor transformation law and in the structure of the spin operators, and it is treated as an abstract algebraic fact about the rotation group.

In this framework the 720 degrees is not abstract. Paper 2 derives, from the geometric structure alone, that every mass-bearing configuration carries a 720-degree signature composed of the inner identity scan (360 degrees, by which the configuration confirms what it is) and the outer temporal scan (360 degrees, by which it searches its surroundings for resolution), the two together forming the full manifold scan (Paper 2, Sections 1 and 4.1; the formal derivation is Paper 2, Appendix B, and the same scans are developed in Section 7 of this paper). The correspondence is therefore direct: the 720-degree rotation that the standard formalism requires of a spin-one-half object is the same 720-degree manifold scan that a mass-bearing configuration executes. What the standard treatment records as an algebraic property of the rotation group, the framework derives as the physical two-scan structure of a configuration that has departed from the photon baseline.

This correspondence also explains why the spinor bookkeeping succeeds without the ontology built on top of it. Paper 2.5 (Section 4) sets out how the four-component Dirac spinor, with its gamma matrices and Clifford algebra, admits four plane-wave solutions, and how the negative-energy solutions — a formal

feature of an equation pushed past its domain — were reified into a literal mirror-world ontology of antimatter. Read through the present framework, the electron and positron are not distinct substances but one geometry with the inside-outside assignment spun in opposite directions (Section 6). The spinor formalism captures the correct 720-degree rotational structure — because that structure is real, and is the manifold scan — while the antimatter ontology layered onto it is the over-extension Paper 2.5 identifies. The framework keeps the bookkeeping and discards the ontology.

What remains to be written explicitly is the operator-level map: the identification of the axis spin with the raising and lowering action of the standard spin operators, carried out term by term to show the operator algebra closes. This framework supplies the geometric process for which that algebra is the shorthand, and the structural correspondence is fixed by the two results cited above; the explicit operator algebra is developed in the series' formal-correspondence work, incorporated into the next paper of the published roadmap rather than asserted here.